Bridging Deep Learning and Gait Physiology

A Neuro-Symbolic Framework
for Reliable Locomotion Classification

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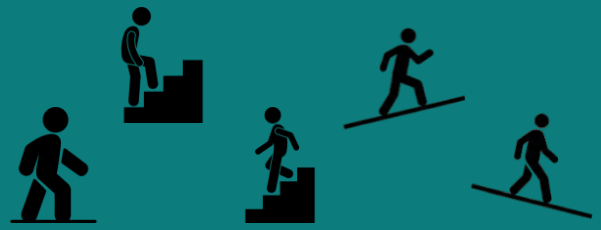


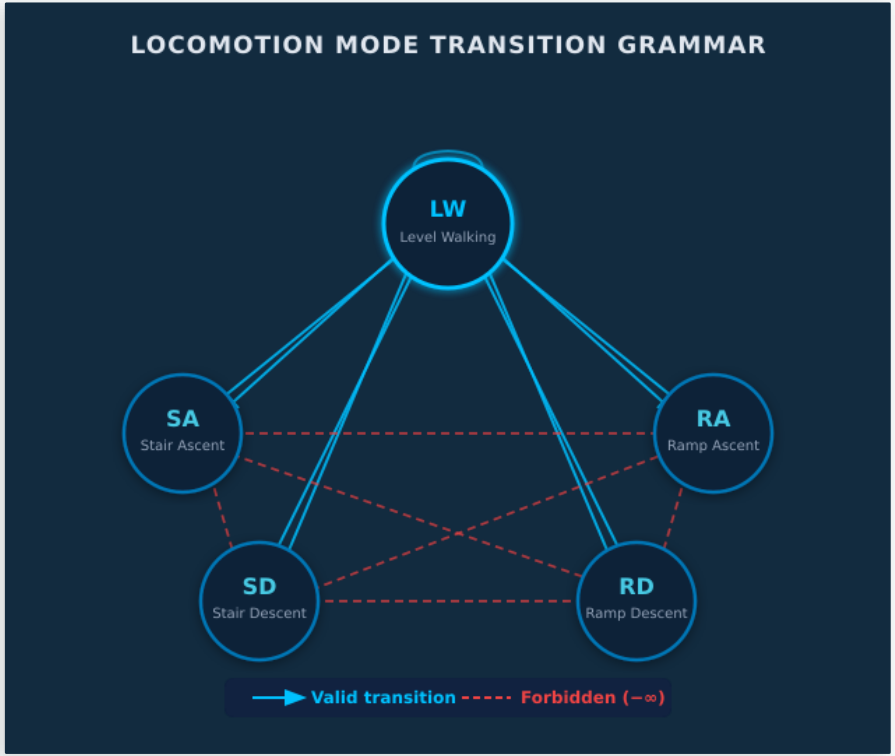
5 Locomotion Modes

- Level Walking (LW)
- Stair Ascent (SA)
- Stair Descent (SD)
- Ramp Ascent (RA)
- Ramp Descent (RD)

Motivation: Why Locomotion Mode Recognition Matters

The Problem

- Lower-limb exoskeletons & orthoses must continuously identify locomotion mode to adapt assistance
- Even brief misclassifications can trigger wrong torque/impedance profiles
- Rapid oscillations in predictions → erratic actuations → safety hazard
- High frame-wise accuracy alone is NOT enough



Exoskeletons rely on LMR for safe torque switching — errors are not just performance issues, they are safety hazards.

The Gap: Frame-wise Accuracy ≠ Temporal Reliability

Forbidden Transitions

SA → RD predicted directly
Biomechanically impossible
Dangerous torque commands

Spurious Oscillations

LW→SA→LW→SA in milliseconds
Physically inconsistent
Erratic actuation

The Gap

Deep models trained with
frame-level loss only
No temporal coherence constraint

Existing Approaches & Their Limitations

HMM / HSMM

Smooth probabilistic transitions

Soft constraints only — cannot hard-guarantee safety

FSM Rules

Hard deterministic rules

Handcrafted thresholds, poor generalization across subjects

Our Approach

Neuro-Symbolic: CNN-LSTM + hard post-filter

Best of both worlds

Dataset & Preprocessing

10	5	28	100 Hz	8×
Subjects (3F / 7M)	Locomotion Modes	Input Channels	Sampling Rate	Transitions per Subject

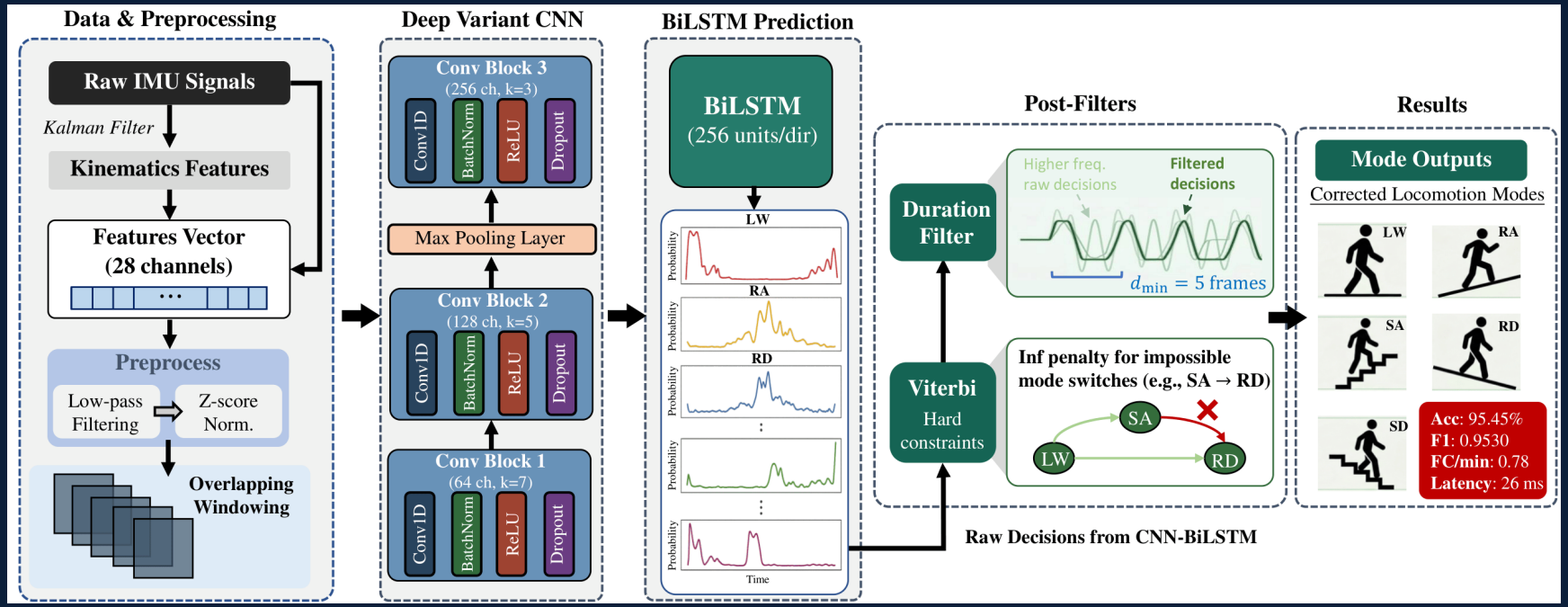
IMU Setup & Features

- 2× foot-mounted IMU (Xsens MTw Awinda)
- Tri-axial accelerometer + gyroscope + quaternion
- Kalman filter → 7 kinematic features (positions, velocities, inclination angles)
- 5-class task: LW, SA, SD, RA, RD

Preprocessing Pipeline

1. **Butterworth LPF**
   4th order, 25 Hz cutoff — removes noise, keeps gait content
2. **Z-score Normalization**
   Fitted on training subjects only — no leakage
3. **Sliding Window**
   W = 2.5 s, overlap = 75% → 250 samples/window
4. **Balanced Class Weights**
   $wc = N / (K \cdot Nc)$ — prevents LW dominance

System Overview: The Two-Stage Pipeline



The post-filter runs entirely on network outputs — lightweight, deterministic, deployable on embedded hardware.

Deep Sequential Classifier: CNN-BiLSTM Architecture

CNN-BiLSTM Deep Variant

- Input · 28-channel, 250 timesteps
- Conv Block 1 · 64 ch, k=7, BatchNorm, ReLU
- Conv Block 2 · 128 ch, k=5, BatchNorm, ReLU
- Max Pooling · Stride 2
- Conv Block 3 · 256 ch, k=3, BatchNorm, ReLU
- BiLSTM · 256 units/dir, 2 layers
- Linear Interp. · Restore full temporal resolution
- FC Head · 5-class output per frame

Why CNN + LSTM?

CNN extracts

Local gait events (heel strike, toe-off) at ms scale

LSTM integrates

Long-range temporal context across gait cycles

Max-pool trick

Reduces LSTM compute by 2× — faster & more efficient

Architecture Comparison (LOSO)

Model	Acc (%)	F1	Time
LSTM baseline	93.28 ± 3.1	0.9245	30.5 min
CNN-LSTM standard	95.89 ± 2.6	0.9575	36.8 min
CNN-LSTM deep ★	**96.11 ± 2.8**	**0.9586**	**12.9 min**
CNN-LSTM light	93.46 ± 3.1	0.9339	3.8 min

Post-Filter Stage 1: Constrained Viterbi Decoding

How Viterbi Works Here

Global Optimality

Instead of greedy argmax at each frame, finds the sequence maximizing total log-probability

Hard Constraint

Forbidden transitions assigned $-\infty$ penalty → physically impossible in output, regardless of NN confidence

Transition Grammar

Only LW↔SA, LW↔SD, LW↔RA, LW↔RD allowed — all stair/ramp direct switches blocked

$O(T \cdot K^2)$ Complexity

Dynamic programming — efficient even for long sequences at 100 Hz

Viterbi Recursion

$$V_t(c) = p_{t,c} + \max_{c'}[\, V_{t-1}(c') + \tau(c',c) \,]$$

$\tau(s,s') = 0$ if allowed | $-\infty$ if forbidden

Result: globally optimal path recovered by backtracking

Transition Mask (✓ = allowed)

	LW	SA	SD	RA	RD
LW	✓	✓	✓	✓	✓
SA	✓	✓	✗	✗	✗
SD	✓	✗	✓	✗	✗
RA	✓	✗	✗	✓	✗
RD	✓	✗	✗	✗	✓

Post-Filter Stage 2: Minimum Duration Filter

The Confirmation Principle

A genuine locomotion mode change requires sustained biomechanical evidence:

→ Foot placement must reposition
→ Center of mass must shift
→ Muscle activation must adapt

The duration filter requires d_min consecutive frames of the new mode before accepting the transition — encoding this physical invariant as a hard algorithmic rule.

Validation-selected d* per subject: 3–7 frames (30–70 ms)

Reflects individual kinematic predictability — adaptive & physically meaningful

d_min Selection Strategy

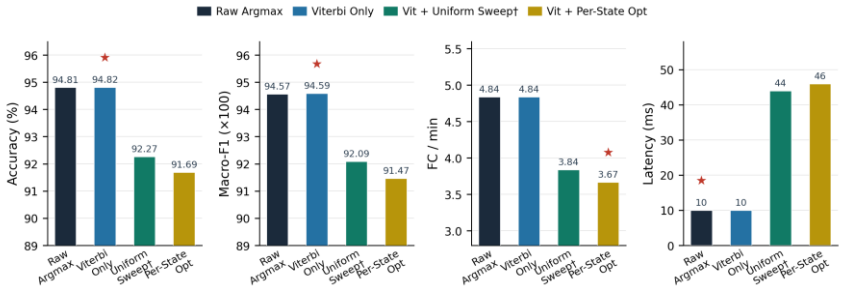
Uniform Sweep

Single d* shared across all modes
Grid: {3,5,7,...,60} frames | ε = 0.05
Objective: min[2·FC/min + Lat/500]

Per-State Optimization

Independent d*(c) per mode — stair/ramp get longer windows
Cartesian product grid | ε = 0.03

The Stability–Responsiveness Trade-off



Evaluation Protocol & Metrics

Strict LOSO Cross-Validation (10 Folds)

No test subject data used at any stage — no leakage

4 Evaluation Metrics

Accuracy	% correctly classified frames
Macro-F1	Balanced across all 5 classes
FC / min	False state changes per minute
Latency (ms)	Median transition detection delay

Ablation Study Highlights — Best Configuration: Window 2.5 s · Overlap 75% · Dropout 0.3

Factor	Best Value	Acc (%)	F1	Insight
Window Size	**2.5 s**	96.11 ± 2.82	0.9586	↑ *full gait cycles captured*
Overlap	**75%**	96.83 ± 2.89	0.9665	↑ *denser training samples*
Dropout	**0.3**	96.01 ± 2.51	0.9583	↑ *best acc–reg balance*

Post-Filtering Results: Key Findings

>20%
FC/min Reduction

10→44ms
Latency Increase (expected trade-off)

92.27%
Accuracy (Uniform Sweep)

3.84
FC/min after filtering

LOSO Mean Results (10 Subjects)

Method	Acc (%)	F1	FC/min	Lat (ms)
Raw argmax	94.81	0.9457	4.84	10
Vit + Uniform ★	**92.27**	**0.9209**	**3.84**	**44**
Vit + Per-state	91.69	0.9147	3.67	46

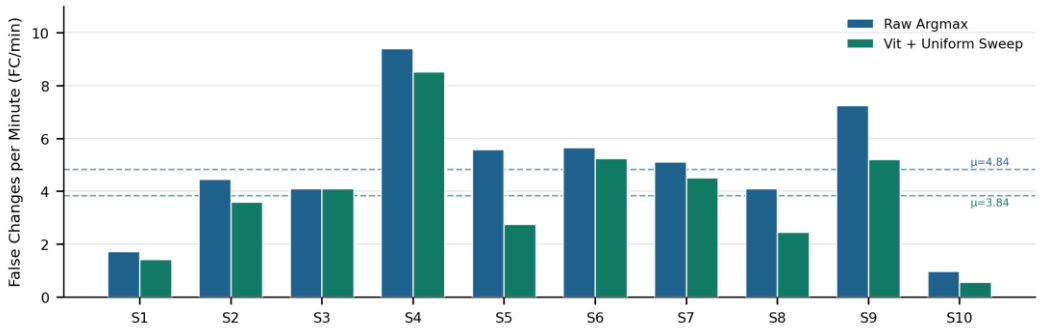
★ Recommended configuration

Key Findings

- Duration filter: 20.7% FC/min reduction — eliminates boundary chattering
- Latency 10→44ms: physically expected — d_min imposes deterministic delay
- Uniform sweep outperforms per-state: more robust with 9 training subjects

Per-Subject Analysis: Two Patterns of Difficulty

FC/min Before vs. After Post-Filtering



Pattern 1 — Low Accuracy

S4 (88.79%), S6 (89.63%), S9 (94.32%)
Structural confusion from atypical gait
Filter cannot recover — commits to wrong state

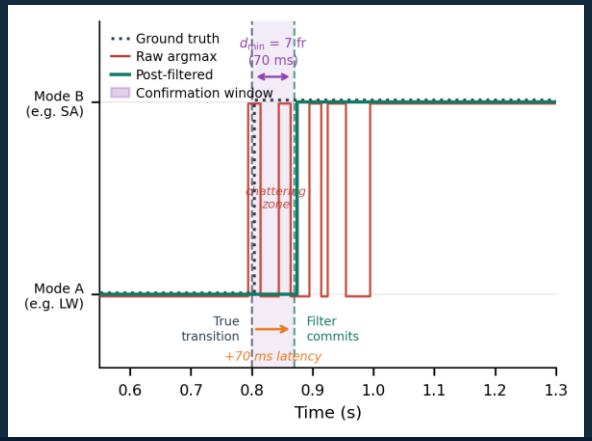
Pattern 2 — High FC/min

S4 (9.41), S9 (7.25), S7 (5.12)
S7: high acc (96.15%) but oscillates
→ local boundary ambiguity only
Filter reduces: S4 9.41→8.53, S9 7.25→5.21

Takeaway: The filter is a stability mechanism for reliable predictions — it cannot fix structural classification errors. High-performing subjects (S1, S8, S10) benefit most: meaningful FC/min reduction with minimal accuracy cost.

The Stability–Responsiveness Trade-off

Conceptual Illustration: Duration Filter Mechanism



What we sacrifice

34 ms extra latency
(10 ms → 44 ms)

What we gain

Zero spurious oscillations
>20% FC/min reduction

The verdict

For exoskeleton control:
stability >> 34 ms

Discussion

Physical Interpretation of the Post-Filter

Viterbi = hard safety bound (not accuracy booster)
Duration filter = algorithmic hysteresis band
Both enforce distinct physical invariants deterministically

Hierarchical Spatiotemporal Representation

CNN: local gait events (heel strike, toe-off) at ms timescale
LSTM: long-range context over gait cycles
Max-pooling: 2× efficiency — redundant 100 Hz resolved by CNN

Filtering Strategy Comparison

Per-state stronger FC/min (24.1%) but 3.29pp accuracy cost exceeds tolerance
Uniform sweep more robust with only 9 training subjects per fold
Per-state needs larger datasets to reliably estimate per-mode statistics

Limitations & Future Work

Able-bodied subjects only → clinical translation needed
Atypical kinematics not covered → subject-specific adaptation
Heuristic d_min → replace with learned HSMM duration distributions
Next: real-time closed-loop exoskeleton validation

Conclusion

Neuro-Symbolic Framework for Safe Locomotion Recognition

1. CNN-BiLSTM deep model achieves 96.11% ± 2.8% accuracy under strict LOSO — +2.8% over LSTM baseline
2. Constrained Viterbi provides hard biomechanical safety guarantee — physically impossible mode sequences eliminated
3. Duration filter reduces false state changes by >20% (4.84 → 3.84 FC/min) by eliminating boundary chattering
4. Latency increases 10→44 ms: deliberate, deterministic, physically expected cost of the stability–responsiveness trade-off
5. No retraining needed — post-filter operates purely on network outputs, deployable on embedded exoskeleton hardware

Future Work: Subject-specific adaptation · Learned HSMM duration models · Real-time closed-loop exoskeleton validation

Thank You

Questions & Discussion

96.11%	>20%	44 ms	100%
Best Accuracy (CNN-LSTM Deep)	FC/min Reduction	Stable Latency (trade-off)	Forbidden Transitions Eliminated

Manel Ouchar · Université Paris-Est Créteil · 2025/2026